



GREENTRUST PULSE

P R E S E N T A T I O N S L I D E

UM Hackathon 2026

Presenter: TriforceTech



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INTRODUCTION



ESG (Environmental, Social, and Governance) is a framework for measuring a company's sustainability and ethical impact.

→ By 2026, it has become a key business metric assessing carbon emissions, labor practices, and data transparency.

ESG Score: The Heart of Our System

A verified digital reputation that turns a factory's sustainability performance into a measurable financial asset, providing a transparent score that proves low risk to lenders.

**** ESG score help to save money on loans and make money from big contracts.**

GreenTrust Pulse is an AI platform that helps Malaysian manufacturing SMEs access green funding by turning fragmented data into a verified ESG credit profile. Using multi-agent reasoning and GraphRAG, it enables factories to detect risks, qualify for low-interest loans, and become trusted partners in global sustainable supply chains.

RESEARCH PAPER



Ant International, IFC, GCash pioneer first-of-its-kind sustainability impact scorecard to expand MSME access to sustainable financing



Manila, Philippines – Furthering its mission of driving digital financial inclusion, Ant International, International Finance Corp. (IFC), and GCash are developing a Sustainability Impact Scorecard that will make it easier for micro, small, and medium enterprises (MSMEs) to track their environmental and social impact, easily adopt sustainability practices in business operations, and align with global standards.

The Scorecard provides Filipino MSMEs with trusted, clear, and accessible pathways to measure sustainability performance, strengthen credibility, and unlock opportunities aligned with global environmental, social, and governance (ESG) standards. By leveraging the GCash ecosystem and metrics tailored to the Philippine market, the Scorecard aims to empower local MSMEs with relevant, simple insights to help transform businesses.

Malaysia ESG Reporting: The Next Chapter, 2026 and Beyond

Introduction

For years, ESG reporting in Malaysia has been evolving quietly in the background. Companies disclosed sustainability information; investors skimmed through it, and regulators encouraged better practices, often without clear consequences for those that fell short. That period of gentle persuasion is coming to an end.

As Malaysia moves toward 2026, ESG reporting is no longer a soft expectation or a reputational exercise. It is becoming structured, mandatory, and closely tied to how companies explain risk, performance, and long-term value. The future involves more than just additional reporting; it signifies a fundamental shift in how sustainability data is compiled, managed, and utilized.

**** Low Carbon Transition Facility (LCTF)** is a specific type of green financing designed for Malaysian SMEs

Low Carbon Transition Facility (LCTF)

Source: International Energy Agency

Last updated: 25 June 2025

In 2022, Bank Negara Malaysia (BNM), the Central Bank of Malaysia, launched a new Low Carbon Transition Facility (LCTF) under BNM's fund for Small Medium Enterprises (SME). LCTF is a financing facility to encourage the commitment of SMEs in all sectors towards low carbon and sustainable business operations through financing capital expenditure.

This includes obtaining sustainability certification, ramping up the use of sustainable materials for production, improving energy efficiency of buildings and appliances as well as installing on-site generation equipment of renewable energy.

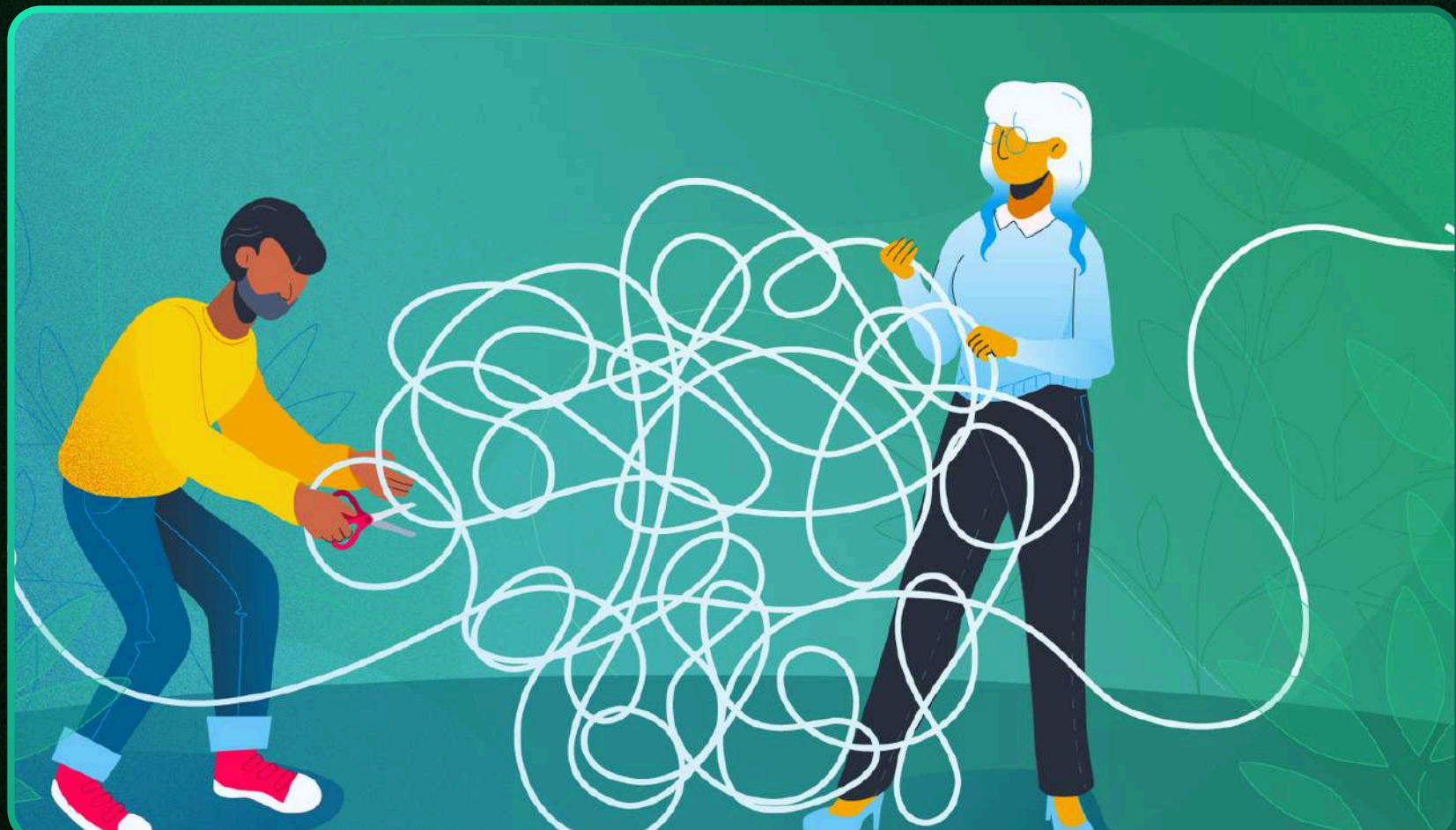
PROBLEM STATEMENT



PROBLEM STATEMENT

► The Financing Gap

Banks offer attractive Green Loans (for renewable energy/green buildings) and ESG Loans (for sustainability-linked targets), but factories lack the structured, "banker-ready" data to prove eligibility.



► The Operational Blindspot

Manufacturing is data-heavy but "intelligence-poor". Fragmented logs and old machinery hide the true ESG impact.

► The Supply Chain Squeeze

MNCs are dropping "dirty" suppliers; without a verified ESG score, factories risk losing major contracts.

OBJECTIVE



01

Improve financing accessibility by helping SMEs qualify for schemes like the Green Technology Financing Scheme (GTFS) with rates as low as 2%, addressing the issue where fewer than 8% currently qualify due to limited visibility and structured reporting.

02

Mitigate operational and commercial risks by detecting supply chain disruptions early and preventing the loss of contracts with ESG-focused multinational corporations (MNCs).

03

Provide explainable intelligence through transparent, auditable ESG scores and recommendations, reducing due diligence efforts and increasing trust for financial institutions.

TARGET USER



Primary User

SMEs

Reduce utility bills, upgrade machinery, and secure the factory's 10-year operational future.



Secondary User

Bank Credit Officers

Require objective, auditable ESG proof to approve lower interest rates without deploying manual, costly site-audits



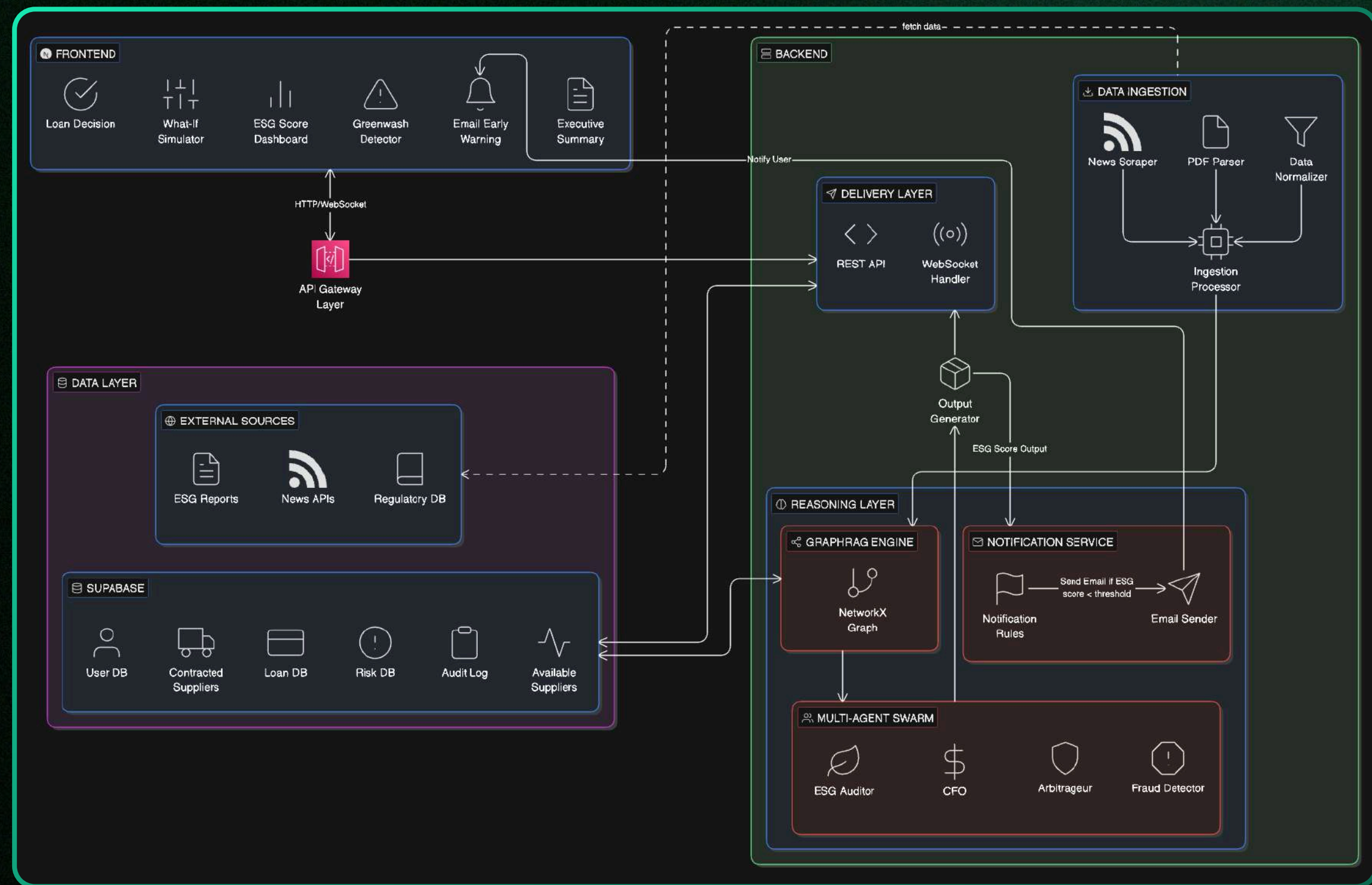
Tertiary User

Ant International

Building a unified system that turns a factory's green efforts into a trusted ESG credit score for banks.



SYSTEM ARCHITECTURE



- Frontend Layer
- Data Layer
- Backend Layer
- API Gateway Layer

TECHNICAL PILLARS



Z.ai GLM Service Layer

An AI-powered, cloud-based decision engine designed to replace static and manual spreadsheets.

24/7 Data Ingestion

Continuous streaming processing that replaces delayed periodic reporting.

GraphRAG Context

A system that instantly links factory operations, external news, and regulatory compliance into actionable insights.

Sequential Multi-Agent Reasoning

Uses four specialized agents: Fraud Detector, ESG Auditor, CFO, and Arbitrageur to cross-check data and resolve trade-offs before delivering a final, explainable loan decision.

CORE PRODUCT FEATURES



Banker-Ready ESG Profile

Delivers a GraphRAG-driven ESG score with automated green loan eligibility checks and personalized financing recommendations, acting like a virtual loan officer.

Integrity Guard & Fraud Detection

Monitors data consistency to detect greenwashing risks and provides a transparent reasoning trace, ensuring trust and auditability.

Predictive Strategy & Simulation

Enables “what-if” simulations to forecast financial and ESG impacts, model strategies for securing long-term MNC contracts.

Real-Time Compliance Pulse

Provides instant alerts on ESG risks and deadlines, along with AI-generated executive summaries for quick reporting to management and banks.

FUTURE IMPLEMENTATION



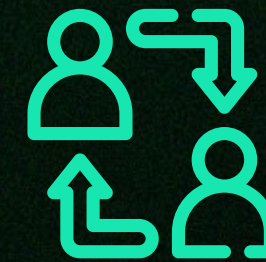
Self-Healing Supply Chain

AI that automatically pivots to "green" suppliers when risks are detected, re-weighting the knowledge graph in real-time.



ESG Leaderboard

An anonymized industry benchmark that gamifies compliance, ranking SMEs against sector peers to drive continuous improvement.



"Regulator" Agent

A meta-auditor agent that runs in parallel to ensure every AI recommendation is 100% compliant with the latest April 2026 mandates.

FEASIBILITY

► Technical Feasibility

- Uses centralized AI reasoning for multi-step ESG decision-making
- Integrates GraphRAG to connect fragmented data into insights
- Four agents cross-validate data for reliable loan decisions



► Market Feasibility

- Targets Malaysia's SMEs, with under 8% qualifying for green financing
- Aligns with GTFS low-interest schemes, driving financing readiness demand
- Helps SMEs meet ESG requirements to secure MNC contracts

TESTING



Two main testing Approach :

▶ PyTest (Backend Automation)

Used for:

- Unit testing (ESG computation, GraphRAG, loan engine)
- Integration testing (multi-module data flow)
- Regression testing (prevent feature breakage)

▶ Postman (API & Performance Testing)

Used for:

- API endpoint validation
- Request/response correctness
- Latency benchmarking under real conditions

MARKET VALUE



Why GreenTrust Pulse Outperforms ?

	Traditional ESG Assessment	GreenTrust Pulse
Data Source	Manual periodic spreadsheet & disjointed PDFs	24/7 continuous API and structured / unstructured ingestion
Risk Detection	Reactive, intuition-based with a trailing 12-month lag	Live GraphRAG supply chain event & news alerts
Verification	Single-point manual consultant trust	Multi-agent Swarm internal cross-validation
Final Output	Static, instantly, outdated, PDF report	Dynamic, banker-ready predictive score profile

MARKET VALUE



Competitive Advantage of GreenTrust Pulse



Bridging the Access Gap

Helping the 92% of SMEs currently failing to access green loans by providing "banker-ready" profiles.



Protecting Supply Chain Contracts

Ensuring SMEs can demonstrate the verifiable ESG performance required to retain high-value contracts with multinational corporations (MNCs).



Cost Efficiency

Replacing manual, high-cost consulting with an automated AI platform to lower implementation barriers for smaller firms.

CONCLUSION

- ▶ Bridges sustainability and finance by transforming fragmented operational data into clear, actionable ESG insights for better decision-making
- ▶ Improves transparency and enhances SMEs' readiness for green financing, increasing their chances of qualifying for funding
- ▶ Provides explainable and auditable intelligence, building trust with financial institutions while simplifying due diligence processes





GREENTRUST PULSE



THANK YOU

FOR YOUR ATTENTION

TEAM

TriforceTech

GITHUB LINK

https://github.com/fciacia/121_TriforceTech.git